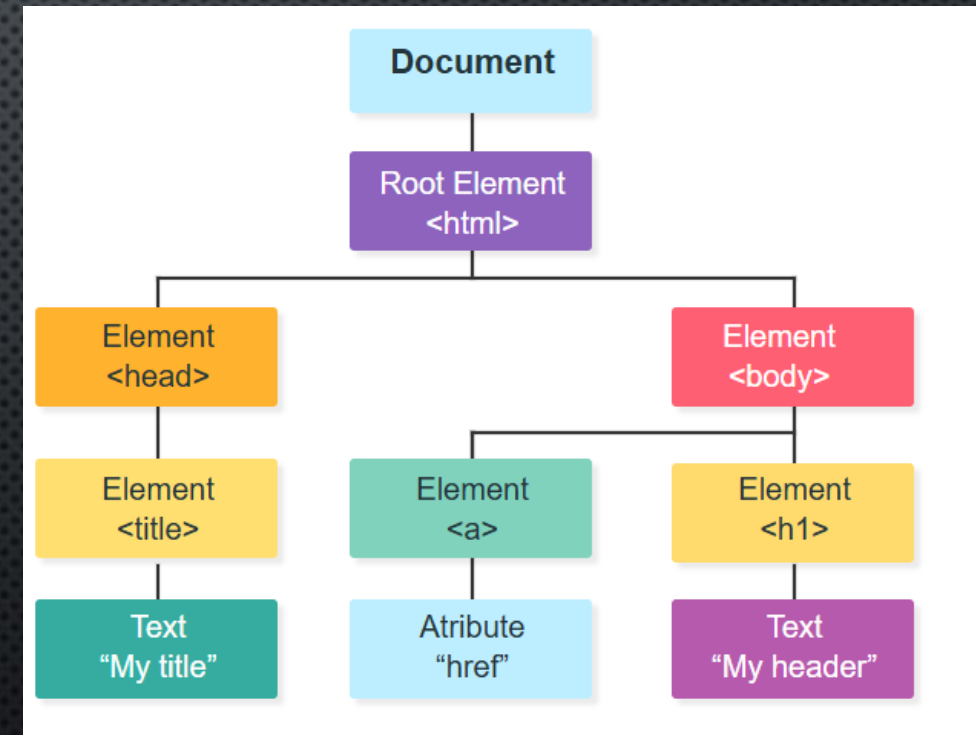


JAVASCRIPT ESSENTIALS

DOCUMENT OBJECT MODEL (DOM)

THE DOM REPRESENTS A DOCUMENT WITH A LOGICAL TREE. EACH BRANCH OF THE TREE ENDS IN A NODE, AND EACH NODE CONTAINS OBJECTS.

DOM METHODS ALLOW PROGRAMMATIC ACCESS TO THE TREE. WITH THEM, YOU CAN CHANGE THE DOCUMENT'S STRUCTURE, STYLE, OR CONTENT.



SELECTING ELEMENTS

Selection methods

- `getElementById()` selects by ID.
- `querySelector()` selects the first match for a CSS selector.
- `querySelectorAll()` returns all matches as a static `NodeList`.
- `getAttribute()` reads any attribute of an element.
- `textContent`, `innerHTML`

DOM TRAVERSAL

- `parentElement`, `children`, `nextElementSibling`, and `previousElementSibling` Navigate between elements

ATTRIBUTES AND CLASSES

Manage HTML attributes

- `getAttribute`, `setAttribute()` and `removeAttribute()`

CSS class manipulation

- `classList` => `add()`, `remove()`, `toggle()`, `replace()`
- and `contains()`

CREATING AND REMOVING ELEMENTS

`document.createElement()` creates a new element. Set its properties and content before inserting it into the DOM.

`appendChild()` adds a child at the end. `insertBefore()` inserts before a reference node. `remove()` detaches an element from the DOM.

EVENT LISTENERS AND EVENT HANDLING

ADDEVENTLISTENER

- `ADDEVENTLISTENER()` ATTACHES A HANDLER FUNCTION TO AN ELEMENT FOR A SPECIFIC EVENT TYPE. IT SUPPORTS MULTIPLE HANDLERS PER EVENT.

EVENT DELEGATION

- INSTEAD OF ATTACHING LISTENERS TO EACH CHILD, ATTACH ONE LISTENER TO THE PARENT AND USE ``EVENT.TARGET`` TO IDENTIFY WHICH CHILD WAS INTERACTED WITH.

SPA (SINGLE PAGE APPLICATION)

A SPA IS A WEB APPLICATION THAT:

- LOADS A SINGLE HTML PAGE
- DYNAMICALLY UPDATES CONTENT WITHOUT RELOADING THE PAGE
- USES JAVASCRIPT TO HANDLE NAVIGATION, RENDERING, AND STATE

NODE JS

FREE, OPEN-SOURCE, AND CROSS-
PLATFORM JAVASCRIPT RUNTIME
ENVIRONMENT THAT LETS DEVELOPERS RUN
JAVASCRIPT CODE OUTSIDE OF A WEB
BROWSER, SPECIFICALLY ON A COMPUTER
OR SERVER

CORE ARCHITECTURE CONCEPTS



V8 Engine



Asynchronous
& Event-Driven



Non-Blocking
I/O



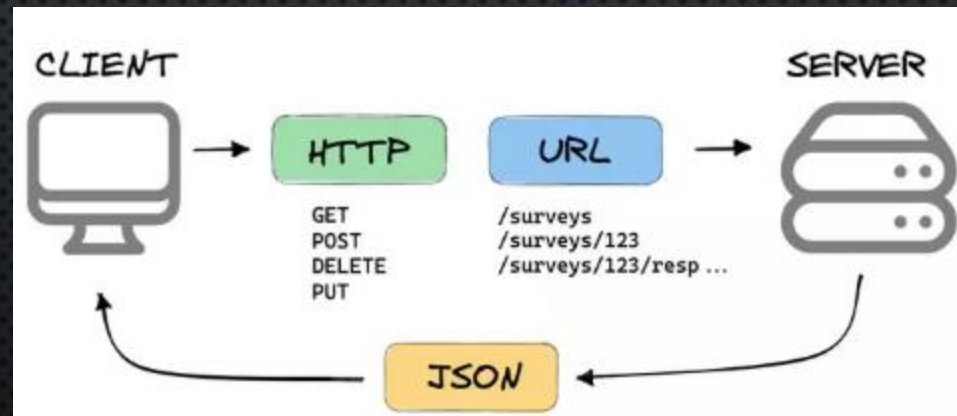
Single-
Threaded



Event Loop

REST (REPRESENTATIONAL STATE TRANSFER)

A REST SERVICE IS A WEB SERVICE THAT ALLOWS DIFFERENT SOFTWARE APPLICATIONS TO COMMUNICATE WITH EACH OTHER OVER THE INTERNET USING STANDARD HTTP METHODS. IT FOCUSES ON RESOURCES (LIKE USERS, PRODUCTS, OR POSTS) IDENTIFIED BY URLS, AND TRANSFERS THE DATA REPRESENTING THOSE RESOURCES IN FORMATS LIKE JSON OR XML



EXPRESS JS

MINIMAL, FLEXIBLE, AND UNOPINIONATED OPEN-SOURCE WEB APPLICATION FRAMEWORK FOR NODE.JS, DESIGNED TO BUILD WEB APPLICATIONS AND APIs QUICKLY AND EASILY

- RESOURCES AND ENDPOINTS
- HTTP METHODS
- STATELESSNESS
- MIDDLEWARE

ENVIRONMENT VARIABLES

- AN ENVIRONMENT VARIABLE IS A KEY-VALUE PAIR THAT LIVES OUTSIDE YOUR CODE, IN THE OPERATING SYSTEM'S PROCESS ENVIRONMENT
- KEEP CONFIGURATION SEPARATE FROM SOURCE CODE
- THE SAME .JS FILES RUN IN DEV, STAGING, AND PRODUCTION, BUT THEY BEHAVE DIFFERENTLY BECAUSE THE ENVIRONMENT TELLS THEM HOW.
- .ENV FILES AND A CONFIG PATTERN
- .GITIGNORE FILE

```
node --env-file=.env ./server.js
```

SUPERTEST

- SUPERTTEST IS A NODE.JS LIBRARY USED FOR TESTING HTTP SERVERS.
- IT PROVIDES A HIGH-LEVEL ABSTRACTION FOR TESTING API ENDPOINTS, ALLOWING YOU TO SIMULATE HTTP REQUESTS WITHOUT MANUALLY STARTING YOUR SERVER DURING TESTS.

```
npm install -D vitest supertest  
npm install -D @vitest/coverage-v8
```

TRABAJO FINAL

- IMPLEMENTAR UN SISTEMA PARA UNA TIENDA DE LIBROS (BACKEND, FRONTEND), SOLO USUARIOS REGISTRADOS PUEDEN REALIZAR OPERACIONES SOBRE LOS LIBROS (CRUD), CUALQUIER PERSONA QUE ENTRA AL SITIO PUEDE BUSCAR Y VER LOS DETALLES DE LOS LIBROS DE LA TIENDA

Usuario

- Nombre completo
- Clave
- Email
- Edad
- Dirección
- Teléfono

Libro

- Título
- Descripción
- Autor
- Categoría
- Idioma
- Año
- Editorial
- Precio
- Moneda

REFERENCES

https://developer.mozilla.org/en-US/docs/Learn_web_development

<https://dev.to/robodobodob/htmx-rethinking-the-spa-2fn2>

<https://www.abbacustechologies.com/the-benefits-of-single-page-applications-spas/>

<https://www.codecademy.com/article/what-is-a-single-page-application-spa>

<https://www.altexsoft.com/blog/mean-mern-javascript-full-stack/>

<https://vitest.dev/guide/learn/writing-tests.html>

<https://expressjs.com>

<https://habr.com/en/articles/1036016/>